

AI Search Framework: Experiment 2 (E2) Empirical Evaluation Report

Query-Dependent SLM Fine-Tuning vs. 4-Tier Non-Fine-Tuned Baseline Ladder

Repository: dixitabhi1/SLM_PROJECT (Branch: exp/mentor-protocol-e1)

Benchmark Cohort: 8 Canonical Two-Domain Queries (100% Taxonomy Coverage)

Evaluation Methodology: Symmetrical Double-Blind Judging (64 trials, 16/tier)

Fairness Status: Pre-Flight Verified (11.85B < P_{Baseline} across all 4 tiers)

1. Executive Summary & Experimental Objectives

Experiment 2 (E2) investigates the empirical impact of Query-Dependent Specialist Fine-Tuning under the Mentor Experiment Protocol. In contrast to Experiment 1 (E1) where all SLM pool models were un-adapted, E2 selectively routes subtasks based on query domain requirements: technical coding/systems subtasks are routed to phi3.5-ft-coding:latest (3.82B parameters, local RTX 3050 GPU), while general and analytical formulation tasks are executed by the base general specialist (meta-llama/llama-3.1-8B-Instruct). All 4 Baselines remain strictly un-adapted and non-fine-tuned, matched directly against cached E1 baseline responses for 100% paired-trial fidelity.

2. Pre-Flight Governance Assertions

- Hard Rule 13 (Distinct Models):** All systems map to distinct checkpoints on separate endpoints (0 collisions).
- Hard Rule 16(b) (Fairness Constraint Pre-Flight):** Combined participating SLM parameters (3.82B + 8.03B = **11.85B**) strictly < 20.0B < 32.0B < 72.7B < 120.0B across all tiers.

3. Overleaf Master Results Table — Experiment 2 (E2)

Baseline Tier	Baseline Model	Params	Framework Mode	SLM Wins	Draws	LLM Wins	Effective SLM Win (Q _S ≥ Q _L)	Quality Proximity [95% CI]	SLM Score	LLM Score	Mean ΔQ [95% CI]	Matched Gain vs E1
Tier 1 (~20B)	openai/gpt-oss-20b	20.0B	1-10 Holistic	3 (18.75%)	1	12	4 (25.0%)	0.6528 [0.5387, 0.7669]	2.06	4.81	-2.75 [-4.06, -1.44]	-0.125
			1-5 Criteria	1 (6.25%)	3	12	4 (25.0%)	62.50% [49.94%, 75.06%]	1.81	3.23	-1.42 [-1.99, -0.85]	-0.312
Tier 2 (~32B)	gemini-2.5-flash	32.0B	1-10 Holistic	1 (6.25%)	0	15	1 (6.25%)	0.5000 [0.4009, 0.5991]	1.94	6.19	-4.25 [-5.46, -3.04]	+0.500
			1-5 Criteria	1 (6.25%)	0	15	1 (6.25%)	47.40% [37.05%, 57.74%]	1.92	3.90	-1.98 [-2.55, -1.41]	+0.167
Tier 3 (~72B)	Qwen/Qwen2.5-72B-Instruct	72.7B	1-10 Holistic	1 (6.25%)	0	15	1 (6.25%)	0.6389 [0.5625, 0.7153]	2.50	5.62	-3.12 [-3.97, -2.28]	+0.625
			1-5 Criteria	1 (6.25%)	0	15	1 (6.25%)	63.54% [55.29%, 71.78%]	2.06	3.48	-1.42 [-1.80, -1.04]	+0.292
Tier 4 (~120B)	openai/gpt-oss-120b	120.0B	1-10 Holistic	0 (0.0%)	0	16	0 (0.0%)	0.4514 [0.3402, 0.5626]	1.56	6.50	-4.94 [-5.94, -3.94]	+0.312
			1-5 Criteria	0 (0.0%)	0	16	0 (0.0%)	40.62% [32.87%, 48.38%]	1.71	4.08	-2.38 [-2.69, -2.06]	-0.083

4. Matched Gain Analysis (E1 vs. E2)

Baseline Tier	E1 Holistic QP	E2 Holistic QP	ΔQP Gain	E1 Mean ΔQ	E2 Mean ΔQ	Matched ΔQ Gain
Tier 1 (~20B)	0.6389	0.6528	+0.0139	-2.62	-2.75	-0.125
Tier 2 (~32B)	0.4722	0.5000	+0.0278	-4.75	-4.25	+0.500
Tier 3 (~72B)	0.5833	0.6389	+0.0556	-3.75	-3.12	+0.625
Tier 4 (~120B)	0.4167	0.4514	+0.0347	-5.25	-4.94	+0.312